

# Troy University - Program at Hanoi

## University of Science and Technology

---

[Code 1]

### FINAL EXAM QUESTIONS - Fall Semester 2025

**Course:** Applied Statistics (MTH 210). **Time:** 120 minutes

**Instruction:** Give answers correct to 3 decimal places.

**Question 1 (10 points)** A data packet is sent through a network that can operate in two modes: The network is in high-load mode with probability 0.35, whereas the network is in normal-load mode with probability 0.65. Packet loss occurs with probability 0.2 when the network is in high-load mode, or with probability 0.05 when the network is in normal-load mode. Let  $H$  be the event that the network is in high-load mode, and  $L$  be the event that a packet is lost.

- a) Compute the probability that a packet is lost,  $\mathbb{P}(L)$ .
- b) Compute the probability that the network is in normal-load mode given that the packet is not lost,  $\mathbb{P}(H^c|L^c)$ .

**Question 2 (10 points)** Khanh has three coins. One coin is fair, one coin is biased so that the probability of obtaining a head is  $\frac{1}{4}$  and the third coin is biased so that the probability of obtaining a head is  $\frac{1}{3}$ . The random variable  $X$  is the number of heads that Khanh obtains when he throws all three coins at the same time. Find the probability mass function (PMF) and the variance of  $X$ .

**Question 3 (10 points)** The length of time in minutes that a customer queues in a bank is a continuous random variable  $T$  with probability density function (PDF)

$$f_T(t) = \begin{cases} \frac{1}{486}(81 - t^2) & \text{if } t \in [0, 9], \\ 0 & \text{otherwise.} \end{cases}$$

Find the conditional probability  $\mathbb{P}(T \leq 6|T \geq 2)$ .

**Question 4 (10 points)** Let  $X$  and  $Y$  be two jointly continuous random variables with joint probability density function (PDF)  $f_{XY}(x, y) = \begin{cases} \frac{1}{4}x^2 + \frac{5}{12}y & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 2, \\ 0 & \text{otherwise.} \end{cases}$

Find the marginal PDF  $f_Y(y)$  of  $Y$ , and the expectation  $\mathbb{E}[XY]$ .

**Question 5 (30 points)** A random sample  $X_1, X_2, X_3, \dots, X_{20}$  is given from a normal distribution with unknown mean  $\mu = \mathbb{E}[X_i]$  and unknown variance  $\text{Var}(X_i) = \sigma^2$ . For the observed sample, the sample mean is  $\bar{X} = 30.66$ , and the sample variance is  $S^2 = 3.604$ .

- a) Find a 99% confidence interval for  $\mu$ .  
b) Find a 95% confidence interval for  $\sigma^2$ .

**Question 6 (15 points)** A random sample of 12 workers from a mobile phone assembly line is selected from a large number of workers. A manager asks each of these workers to assemble a phone at their normal working speed. The time taken, in minutes, to complete these tasks are recorded below.

43.2, 41.6, 49.3, 48.2, 44.2, 40.6, 39.7, 43.4, 44.9, 45.1, 46.2, 43.2

Assuming that this sample comes from a normal distributed population, test the claim at the 5% significance level that the population mean is different from 45 minutes.

**Question 7 (15 points)** Let  $X_1, X_2, X_3, \dots, X_{100}$  be a random sample from an unknown distribution. After observing this sample, the sample mean is  $\bar{X} = 19.3$ , and the sample variance is  $S^2 = 1.817$ . Design a significance level 0.05 test to choose between  $H_0 : \mu = 20$ ,  $H_1 : \mu < 20$ .

---

The critical values of standard normal distribution are  $z_{0.05} = 1.645$ ,  $z_{0.025} = 1.96$ . The critical values of Student's t-distribution are  $t_{0.005,19} = 2.861$ ,  $t_{0.01,19} = 2.539$ ,  $t_{0.025,11} = 2.201$ ,  $t_{0.05,11} = 1.796$ . The critical values of Chi-Square distribution are  $\chi^2_{0.025,19} = 32.852$ ,  $\chi^2_{0.975,19} = 8.907$ .